Sensor reference — Sprintek SK8707-01-004

The part in use is the SK8707-01-004: 3.3 V, 2.4 × 2.4 × 2.4 mm square prism stick, detached sensor PCB without metal plate. Source: Sprintek datasheet DS0048 ver. 1.04 (March 30, 2026), pages 3 and 6.

Dimensions that drive the model (mm)

Item	Value	Where it is used
Sensor PCB width × length	13.20 × 18.29	placeholder box
Sensor PCB thickness	0.80	placeholder box
Stem cross-section	2.40 × 2.40 square	placeholder stem; keycap/cap socket is square
Stem height above sensor PCB top face	2.40	stem length
Stem center from the castellated-pad edge	10.84	positions the box relative to the stem
Stem center from either side edge	6.60	centered in width
Sensor PCB mounting holes	4 × Ø1.60 on a 9.50 × 9.50 square around the stem	not used by the keyboard footprint (center Ø7.5 hole only)
Castellated contacts	4, at the bottom edge	land on the 4 back-side pads of footprint `REF**` (11.8 mm from center in KiCad)

How it mounts in this board

The KiCad footprint `TP_Sprintek_SK8707-01_Sensor_reversible` has a Ø7.5 mm non-plated hole at its center and four 2.5 × 2.0 mm pads on the back copper layer. So the sensor PCB sits under the keyboard PCB with its stem pointing up through the hole, and its castellations solder to those back-side pads.

Stem top above the keyboard PCB top = 2.40 (stem) − 1.60 (board) = **0.80 mm**, before copper thickness and solder sink. The cap therefore has to reach down into the Ø7.5 hole to engage the stem; `cap_offset` and the cap's socket depth decide the final cap height (open question on the Overview tab).

VarSet defaults (`Params`)

Property	Default	Note
`sensor_pcb_w`	13.20	
`sensor_pcb_l`	18.29	
`sensor_pcb_t`	0.80	
`sensor_stem_size`	2.40	square
`sensor_stem_h`	2.40	above sensor PCB top face
`sensor_stem_from_pad_edge`	10.84	
`cap_offset`	0	socket floor rests on stem top

Other variants (for reference)

-001/-002 integrated prism stick (5 V / 3.3 V); -003 detached with metal plate; -005/-006/-007 M2 screw stick 11.5 / 9.5 / 16.5 mm; -008 9.9 mm screw stick with diamond plate; -010 11.5 mm screw stick with plate. A screw-stick variant would raise the stem top well above the keycaps and is a phase 2 option if the gap work needs a taller cap.

Model (built 2026-10-09)

`sensor_sk8707.py` builds an `App::Part` named `Sensor SK8707-01-004`; `fcstd_colors.py` writes the colors into the saved file. Local frame: origin at the stem center on the PCB top face, +Z up the stem, the castellated edge at −Y. Built size: 13.20 × 18.29 mm, z −0.80 to +2.40.

Object	Built from	Color
Sensor PCB	box minus 4 × Ø1.6 holes on a 9.5 mm square	PCB green (0.11, 0.42, 0.26)
Sensor Stem	2.4 mm square box, 2.4 mm tall	white (0.93, 0.93, 0.90)
Sensor Pad 1–4	1.0 × 1.2 mm plating boxes at −1.5/−0.5/0.5/1.5 × 2.45 mm pitch on the −Y edge	gold (0.83, 0.66, 0.20)
Sensor Ring	Ø7.0 / Ø4.4 annulus, 0.05 mm, around the stem	gold

The pad pitch (2.45 mm) follows the KiCad footprint's back-side pads rather than the datasheet drawing, so the model lands on the pads it solders to.

KiCad footprint (corrected 2026-10-10)

`TP_Sprintek_SK8707-01-004.kicad_mod` replaces the earlier reversible footprint. Receivers from the datasheet, the Sprintek gerbers (2.13 × 2.84 mm on a 2.49 mm pitch, centered on the sensor's edge) and the sensor model.

Item	Old	Corrected
Signal pads 1–4 (B.Cu)	2.5 × 2.0 at x ±1.225 / ±3.675, y 11.8 (overlapping, outside the sensor edge)	2.1 × 2.8 at x −3.75, −1.25, 1.25, 3.75, y 10.84 (centered on the castellated edge)
Corner receivers 5–6	polygons at x ±6.1, y 8.5–11.5 with notch at y 10	2.15 × 3.0 pads at (±6.525, 9.09) straddling the side edges, on the R0.75 half-holes
Outline (Dwgs)	y −7.4 … 11.8	13.2 × 18.29: y −7.45 … 10.84
Center hole	Ø7.5 NPTH	unchanged
3D model	SK8707-01-004.step at z −1.67	z −1.6 (sensor top face against the board underside), rotation 0